

COMPLETE SYSTEM GUIDE

Public Ship Log

User Manual, Architecture Guide & Standard Operating Procedure — a public daily build streak from Telegram notes and GitHub commits, with AI-drafted posts you approve, written in plain English for anyone to follow.

System Architecture

Daily Usage Guide

Making Changes

Maintenance & Security

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1 What Is This System?

In one sentence: **you tell a Telegram bot what you worked on, it also counts your GitHub commits, keeps a daily streak on a public web page, and every night writes a short post that goes to Bluesky only if you approve it.**

What it does

- **Logs your work:** send `/log fixed the login bug 1.5h` . Hours are optional.
- **Counts commits too:** commits you push to the repos listed in `config.json` are added on their own. Bot and CI commits don't count.
- **Keeps a streak in India time (IST):** any day with at least one entry counts. The streak stays alive until midnight IST.
- **Reminds you at 9 PM IST** if nothing is logged yet and your streak is at risk.
- **Drafts a post at 10:30 PM IST:** AI turns the day's notes and commits into a short update, e.g. *"Day 12 of shipping daily 📦 ..."*. You get `Approve & post` , `Skip` and `Rewrite` buttons.
- **Shows it publicly:** a page with your streak, a 12-month heatmap and a timeline of every day.

Where it runs

	Online (normal)	On your PC (testing)
Runs on	GitHub Actions, every 15 minutes	<code>C:\Python314\python dev.py</code>
Page	<code>swapnil-dhavate.github.io/public-ship-log</code>	<code>http://localhost:8107</code>
Bot replies	Within about 15 minutes	Straight away

Current status (4 October 2026)

Built and tested offline (10 tests pass). Not live yet: it still needs its own Telegram bot token, Gemini and Groq keys, the list of repos to track, Bluesky details, and GitHub Pages switched on. The checklist is in Section 7.1.

What it costs

Nothing: ₹0. Every service used has a free tier, and GitHub Actions and Pages are free for public repos.

2 How the Streak Is Counted

The rule is simple: **a day counts if it has at least one entry, and the streak is the number of days in a row, counted in India Standard Time.**

The rules

- 1 One entry is enough.** A `/log` message or one commit makes the day count. Five entries in one day still count as one day.
- 2 Days are IST days.** GitHub stores commit times in UTC. The app moves them to IST first, so a commit at 19:00 UTC on 3 Oct counts for **4 Oct** (00:30 IST).
- 3 Today is a grace day.** If you haven't logged yet today, yesterday's streak is still shown, marked "at risk", until midnight IST.
- 4 A missed day resets it.** The current streak goes back to 0. Your longest streak is kept.
- 5 Only real work counts.** Commits from bots (names ending in `[bot]`), GitHub Actions, Dependabot), merge commits and messages with `[skip ci]` are ignored.

A worked example (today is 4 October)

Date	Entries	Counts?
1 Oct	none	✗ gap
2 Oct	<code>/log wrote the streak tests 2h</code>	✓
3 Oct	2 commits to a tracked repo	✓ (one day)
4 Oct, 8 PM	nothing yet	grace day

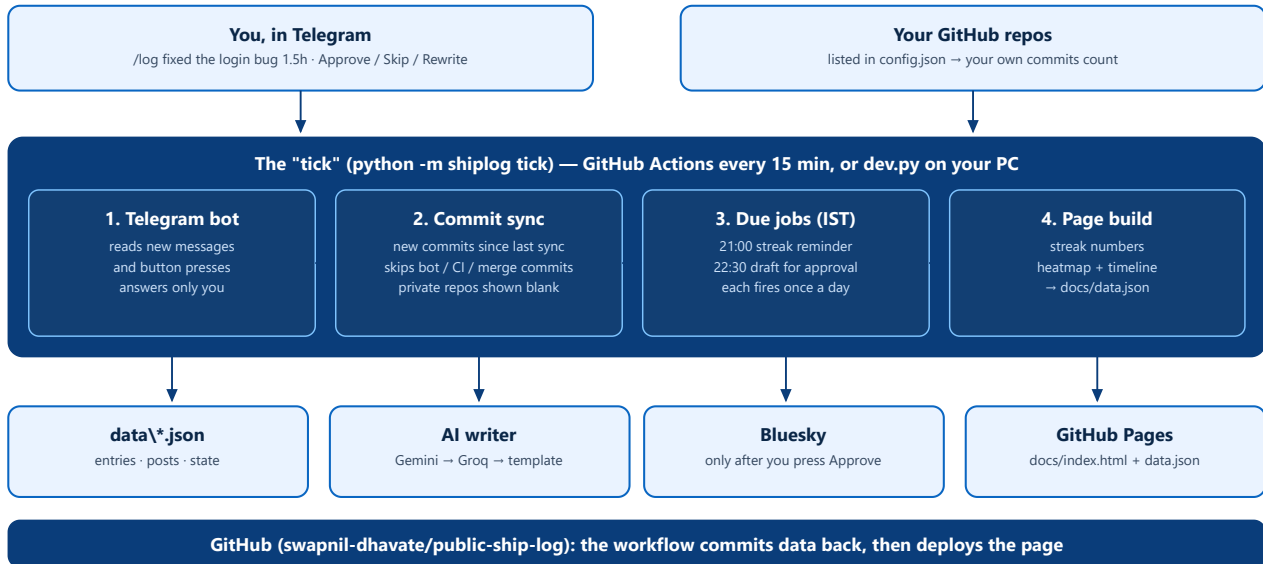
At 8 PM the page shows a **2-day streak, at risk**. At 9 PM IST the bot sends "*Streak at risk!*". If you send `/log fixed the heatmap colours 1h` before midnight, the reply is "✅ *Logged (1h) ... 🔥 Day 3 — streak kept alive!*".

Hours are optional

`2h`, `1.5 hours` and `45m` are read as hours (0.75 for 45m). Numbers that aren't hours, like "read 3 chapters", stay in the text.

3 System Architecture

Every piece involved and how they connect. Each tool is explained in Section 6, and every technical word is in the Glossary.



How a note or a commit becomes a streak, a heatmap square and (after approval) a post.

In words

1. You send messages to the bot and push code to your repos as usual.
2. Every 15 minutes GitHub Actions starts a small Python job, the **tick**. It reads new Telegram messages, pulls new commits, sends the reminder or draft if it's time, and rebuilds the page data.
3. Everything is saved as JSON files in `data\`. The job commits them back to the repo, so the repo itself is the database.
4. Drafts are written by **Gemini**; if that fails, **Groq**; if both fail, a plain template.
5. When you press Approve, the post goes to **Bluesky**. The page is redeployed to **GitHub Pages** whenever the data changed.

4 How It Works, Step by Step

4.1 — One tick (every 15 minutes)

- 1 **Telegram:** fetch messages and button presses since the last one handled (the position is saved in `data\state.json`).
- 2 **Commits:** ask GitHub for commits since the last sync, minus 2 days of overlap to catch late pushes. Already-known commits are skipped.
- 3 **Jobs:** after 21:00 IST with nothing logged → reminder (once a day). After 22:30 IST with something logged and no draft yet → draft.
- 4 **Page:** rebuild `docs\data.json`. If anything changed, commit and deploy the page.

4.2 — Writing the draft

- 1 The day's notes and commits are listed for the AI, with the total hours.
- 2 The AI is told: one update, at most 200 characters, first person, concrete, no hashtags, no hype words.
- 3 The app adds "*Day N of shipping daily* 🇮🇳" in front, then `#buildinpublic` and the page link if they fit in Bluesky's 300-character limit.

4.3 — Approving it

Button	What happens
 Approve & post	Posts to Bluesky and shows the link. In dry-run it is only marked "approved".
 Skip	Marked skipped. Nothing is posted. The day still counts in the streak.
 Rewrite	Asks the AI again and replaces the text in the same message.

If posting fails, the bot says why and the draft stays pending, so you can press Approve again.

5 Layers & Who Does What

Layer	File	Its job
1. Commands	<code>shiplog__main__.py</code>	<code>tick</code> , <code>sync</code> , <code>draft</code> , <code>status</code> and other commands.
2. Bot	<code>shiplog\bot.py</code>	Telegram commands, hours parsing, draft buttons, publishing.
3. Telegram client	<code>shiplog\tg.py</code>	Send, edit and fetch Telegram messages.
4. Streak	<code>shiplog\streak.py</code>	IST clock and the streak numbers.
5. Commits	<code>shiplog\github_sync.py</code>	Pulls commits, skips bots, hides private repo details.
6. Writer	<code>shiplog\writer.py</code>	Gemini → Groq → template; adds day number, tags, link.
7. Publisher	<code>shiplog\bluesky.py</code>	Logs in with the app password, posts, makes links and tags clickable.
8. Timed jobs	<code>shiplog\jobs.py</code>	The 9 PM reminder and the 10:30 PM draft.
9. Storage	<code>shiplog\store.py</code>	Reads and safely writes the JSON files in <code>data\</code> .
10. Page data	<code>shiplog\site.py</code>	Builds <code>docs\data.json</code> ; rewrites it only when something real changed.
11. Settings	<code>shiplog\env.py</code> , <code>config.json</code> , <code>.env</code>	Where every setting and secret comes from.
12. Public page	<code>docs\index.html</code>	Streak, heatmap and timeline; reads <code>data.json</code> .
13. Local mode	<code>dev.py</code>	Page on port 8107 + live bot + jobs on your PC.
14. Schedule	<code>.github\workflows\shiplog.yml</code>	Runs the tick every 15 min, commits data, deploys the page.
15. Checks	<code>tests\test_core.py</code>	10 automatic tests.

6 Tools Used & Their Cost

Tool	Used for	Cost
Python 3 FREE	All the code, standard library only (3.14 on the PC, 3.12 on Actions)	Open-source
Telegram Bot API FREE	Logging, reminders, approval buttons	Free
GitHub REST API FREE	Reading your commits	Free
GitHub Actions FREE	Running the tick every 15 minutes	Free for public repos
GitHub Pages FREE	Hosting the public page	Free for public repos
Gemini (gemini-2.5-flash) FREE	Writing drafts	Free tier
Groq (llama-3.3-70b-versatile) FREE	Backup draft writer	Free tier
Bluesky FREE	Publishing approved posts	Free

Total: ₹0 a month

One draft a day uses a tiny part of the free AI limits. No card is attached to any of these services.

Why these choices?

- **No packages:** nothing to install or update, and nothing breaks when a library changes.
- **GitHub Actions instead of a server:** free and always on. The cost is a slower bot (up to 15 minutes).
- **JSON in the repo instead of a database:** the log is public anyway, and git keeps the full history.
- **Bluesky:** its API is free and allows posting with an app password. X and LinkedIn don't offer that for free.

7 Daily Usage Guide (SOP)

7.1 — First-time setup checklist (still pending)

#	Item	Where it goes
1	New Telegram bot: @BotFather → <code>/newbot</code> → copy the token	<code>TELEGRAM_BOT_TOKEN</code>
2	Your chat id: run <code>dev.py</code> , message the bot, it replies with the id	<code>TELEGRAM_CHAT_ID</code>
3	Gemini key (aistudio.google.com/apikey) and Groq key (console.groq.com/keys)	<code>GEMINI_API_KEY</code> , <code>GROQ_API_KEY</code>
4	Repos to track and your GitHub username	<code>config.json</code> : <code>github_repos</code> , <code>github_author</code>
5	Bluesky handle + app password (Settings → Privacy and security → App passwords)	<code>BLUESKY_HANDLE</code> , <code>BLUESKY_APP_PASSWORD</code>
6	Optional: read-only fine-grained token, only for private repos	<code>GH_TOKEN</code>
7	Copy items 1–6 into repo Settings → Secrets and variables → Actions	Secrets
8	Repo Settings → Pages → Source: GitHub Actions	GitHub
9	Actions → ship-log → Run workflow once; then stop running <code>dev.py</code>	GitHub
10	When happy: variable <code>BLUESKY_DRY_RUN</code> = <code>0</code> ; set <code>site_url</code>	Variables, <code>config.json</code>

For testing on the PC, items 1–6 go in a `.env` file (copy `.env.example`).

7.2 — Every day

- 1 Work on something. Push commits as usual; they count on their own.
- 2 For work without commits (design, reading, writing), send `/log what you did 1h`.
- 3 At 9 PM IST, a reminder comes only if nothing is logged.
- 4 At 10:30 PM IST, read the draft. Press `Approve & post`, `Rewrite` or `Skip`.

Handy commands in Telegram

`/streak` numbers · `/today` what's logged · `/undo` remove your last `/log` · `/draft` write today's post now.

8 How to Make Changes

Most changes are settings in `config.json`. Every change follows the same steps: **edit, test, then push to GitHub**. The next tick uses the new version. You can also just ask Claude to do the edit.

8.1 — Common changes

To change...	Edit this
Page heading and text	<code>title</code> , <code>subtitle</code> in <code>config.json</code>
Repos that count	<code>github_repos</code> , e.g. <code>["swapnil-dhavate/some-repo"]</code>
Reminder and draft times	<code>reminder_time_ist</code> (21:00), <code>draft_time_ist</code> (22:30)
Hashtags and page link in posts	<code>hashtags</code> , <code>site_url</code>
Profile links on the page	<code>links</code> : <code>[{"label": "Bluesky", "url": "..."}]</code>
Show private repo commit messages	<code>redact_private_repos</code> : <code>false</code>
Commit messages to ignore	<code>ignore_commits_matching</code>
Tone of the posts	<code>SYSTEM</code> text in <code>shiplog\writer.py</code>
Page look	<code>docs\index.html</code>

8.2 — Check before publishing

In a terminal, in the project folder:

```
C:\Python314\python -m unittest discover -s tests -t .
```

It should end with **"Ran 10 tests ... OK"**. To see the page: `C:\Python314\python dev.py --page-only` → `http://localhost:8107`.

8.3 — Publish

Open VS Code → `Terminal → New Terminal`, paste this line and press Enter:

```
cd "D:\Swapnil\Personel Project\Weekend AI Projects\09-public-ship-log"; git pull --rebase; git add -A; git commit -m "describe the change"; git push
```

Always pull first

The workflow commits data to the repo every time something changes. `git pull --rebase` brings those commits down first, so your push isn't rejected. Never edit `data\*.json` while Actions is running unless you pull right before and push right after.

9 Adding Features & Growing

9.1 — Track another repo

Add `"owner/repo"` to `github_repos` in `config.json` and push. The very first sync looks back `github_backfill_days` (14) days. After that, each sync looks back only 2 days before the previous one, so a repo added later doesn't get its older history.

9.2 — Go live on Bluesky

- 1 Test with `BLUESKY_DRY_RUN=1` (the default): Approve only marks the draft "approved".
- 2 When the drafts look right, add the Actions variable `BLUESKY_DRY_RUN = 0`.
- 3 Set `site_url` to the Pages address so posts link to your heatmap.

9.3 — Post without asking (optional)

Set `approval_required` to `false`. The nightly draft is then posted straight away and you only get a copy. Keep it `true` until you trust the drafts.

9.4 — Backfill a missed draft

`C:\Python314\python -m shiplog draft --date 2026-10-03` writes and sends a draft for that IST date.

9.5 — Ideas for later

- Post to Mastodon or LinkedIn as well as Bluesky.
- A weekly recap post every Sunday.
- A webhook (for example a free Cloudflare Worker) so the bot replies instantly instead of every 15 minutes.
- Tag entries by project and filter the heatmap.
- An `/edit` command to fix a logged entry.

10 Renewing & Maintaining

Short answer: **very little**. Nothing is paid. These are the things that can expire or change.

Item	Lifetime	What to do
Telegram bot token	Doesn't expire	If it leaks: @BotFather → <code>/revoke</code> , then update the secret.
Gemini / Groq keys	Until revoked	Replace the secret if drafts start saying <code>template</code> .
AI model names	Providers retire models	Set <code>GEMINI_MODEL</code> / <code>GROQ_MODEL</code> to a current model.
Bluesky app password	Until revoked	Make a new one in Bluesky settings, update the secret.
<code>GH_TOKEN</code> (if used)	Fine-grained tokens expire (you pick, max 1 year)	Make a new one before it expires and update the secret.
Scheduled workflow	GitHub pauses schedules after 60 days with no repo activity	Data commits usually keep it active. If paused: Actions → ship-log → <code>Enable workflow</code> .
Data files	Grow slowly	A year of daily use is still small. The page shows the latest 300 entries.

Weekly check (2 minutes)

- 1 Repo → **Actions**: recent runs are green.
- 2 Open the page: today's square and the streak look right.
- 3 In Telegram, `/streak` answers within 15 minutes.

Where secrets live

Online: repo Settings → Secrets and variables → Actions. On the PC: the `.env` file, which git ignores. Nowhere else.

11 Security & Privacy

What is public on purpose

- **The page and the whole repo**, including `data\entries.json` (every note and commit message) and `data\posts.json` (every draft, including skipped ones).
- **Actions run logs** are visible to anyone on a public repo. The code never prints keys, but error messages from services can appear there.

Don't log secrets or employer details

Anything you send with `/log` becomes public. Keep notes about side projects only.

What stays private

- **Keys and tokens** are in GitHub Secrets or the git-ignored `.env`, never in the code.
- **Private repos**: their commits appear as "Pushed a commit to a private project", with no name or message (`redact_private_repos = true`).
- **Bluesky** uses an app password, which can be revoked without touching your real password.

Who can control the bot

- Only the chat in `TELEGRAM_CHAT_ID`. Messages and button presses from anyone else are ignored silently.
- Until `TELEGRAM_CHAT_ID` is set, the bot replies to anyone with their own chat id and does nothing else. Set it straight away.

Nothing is posted by surprise

Guard	Effect
<code>approval_required = true</code>	Every post waits for your Approve press
<code>BLUESKY_DRY_RUN = 1</code> (default)	Even Approve doesn't really post
Bluesky not configured	Approve only marks the draft "approved"

12

Technical Reference

Addresses

Public page (after Pages is on)	<code>https://swapnil-dhavate.github.io/public-ship-log/</code>
Code	<code>https://github.com/swapnil-dhavate/public-ship-log</code>
Local page	<code>http://localhost:8107</code> (<code>dev.py</code>)
Project folder	<code>D:\Swapnil\Personel Project\Weekend AI Projects\09-public-ship-log</code>

Commands (`C:\Python314\python -m shiplog ...`)

<code>tick</code>	Telegram + commits + due jobs + page data (what Actions runs)
<code>poll</code> / <code>sync</code>	Only Telegram / only commits
<code>draft</code> [<code>--date YYYY-MM-DD</code>]	Write a draft and send it for approval
<code>remind</code>	Run the reminder/draft checks (fire only when due)
<code>build</code> / <code>status</code>	Rebuild <code>docs\data.json</code> / print streak numbers
<code>chatid</code>	List chat ids that messaged the bot

`dev.py` options: `--no-schedule` (no reminder or draft), `--page-only` (no bot). It checks commits every 10 minutes.

Settings in .env / Actions

Secrets	<code>TELEGRAM_BOT_TOKEN</code> , <code>TELEGRAM_CHAT_ID</code> , <code>GEMINI_API_KEY</code> , <code>GROQ_API_KEY</code> , <code>GH_TOKEN</code> , <code>BLUESKY_HANDLE</code> , <code>BLUESKY_APP_PASSWORD</code>
Optional	<code>GEMINI_MODEL</code> , <code>GROQ_MODEL</code> , <code>BLUESKY_PDS</code> (default <code>https://bsky.social</code>)
Variable	<code>BLUESKY_DRY_RUN</code> (defaults to 1 if unset)

Workflow facts

- Cron `*/15 * * * *` plus a manual `Run workflow` . Runs never overlap.
- Commits `data` and `docs/data.json` as *ship-log-bot*, then deploys the whole `docs\` folder to Pages. This manual (`docs\manual\`) is published with it.

13 Troubleshooting & FAQ

The bot doesn't answer

Online it answers within about 15 minutes, and GitHub's timer can run late. Check Actions for a red run. On the PC, `dev.py` must be running and show *"Bot: listening"*.

Some messages get answered, others vanish

`dev.py` and Actions are both reading the same bot. Stop `dev.py`, or use `dev.py --page-only`.

Bot says "Your chat id is ..." to every message

`TELEGRAM_CHAT_ID` isn't set. Put that number in `.env` (and the Actions secret) and restart.

My commits don't show up

Check the repo is in `github_repos` as `owner/repo` and `github_author` is your GitHub username. Private repos need `GH_TOKEN`. Merge commits and `[skip ci]` commits never count. Run `C:\Python314\python -m shiplog sync` to see what it finds.

A commit counted for the wrong day

Days are IST. A commit after midnight IST counts for the next day, even if it was still evening in UTC.

The draft says "template" instead of gemini or groq

Both AI services failed (wrong key, limit reached, or retired model). The template draft still works. Check the keys and model names.

Approve says "Approved (not posted...)"

Dry-run is on or Bluesky isn't set up. Set `BLUESKY_DRY_RUN = 0` and both Bluesky secrets.

"Bluesky post failed"

Usually a wrong handle or app password. The draft stays pending; fix the secret and press Approve again.

The page shows "Couldn't load data.json"

Open it through `http://localhost:8107` or the Pages address, not by double-clicking the file.

14 Known Limits

Limit	Why	Workaround
Not live yet	Keys, bot, repos and Pages still pending	Setup checklist, Section 7.1
Only tested offline	No real Telegram, AI or Bluesky call made yet	Start with dry-run on; watch the first few days
Bot replies are slow online	Actions polls every 15 min, sometimes later	Use <code>dev.py</code> for instant replies (not at the same time)
Late runs can skip a draft	If no tick runs between 22:30 and midnight IST, the job never sees that day	<code>shiplog draft --date ...</code> , or <code>/draft</code> before midnight
Can't run bot in two places	Telegram hands each message to whoever asks first	One place at a time
No way to edit entries in chat	Only <code>/undo</code> (last /log) exists	Edit <code>data\entries.json</code> , then push
Everything logged is public	The repo and page are public by design	Log side-project work only
One user only	One chat id, one author	Fork the repo for another person
Posts to Bluesky only	Other networks not built	Copy the draft by hand
Up to 500 commits per repo per sync	5 pages of 100	Fine for normal use
Times are IST only	Built for India	Change <code>IST</code> in <code>streak.py</code>

Term	Plain-English meaning
Streak	How many days in a row you've logged something.
IST	India Standard Time (UTC + 5:30). The app's clock.
Entry	One logged item: a <code>/log</code> message or one commit.
Commit	A saved change in a GitHub repo.
Repo	A project's folder on GitHub, with its full history.
Tick	One run of the job: read Telegram, get commits, do due jobs, rebuild the page.
GitHub Actions	GitHub's free robot that runs the tick every 15 minutes.
GitHub Pages	GitHub's free website hosting, used for the public page.
Heatmap	The calendar grid. Darker squares mean more activity that day.
Draft	A post written by AI that waits for your decision.
Dry-run	Practice mode: Approve works, but nothing is really posted.
Bluesky / app password	A free social network, and a spare password just for this app that you can cancel any time.
Gemini / Groq	Free AI services that write the drafts. Groq is the backup.
Secret	A key or password stored safely in GitHub settings, never in the code.
Chat id	The number Telegram uses for your chat. The bot only obeys this one.
JSON	A plain text format for data. All the log data is kept in JSON files.
localhost	"This computer". <code>localhost:8107</code> only opens on your PC.

Quick Reference Card

Addresses

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Code	<code>https://github.com/swapnil-dhavate/public-ship-log</code>
Local page + bot	<code>C:\Python314\python dev.py</code> → <code>http://localhost:8107</code>
Project folder	<code>D:\Swapnil\Personel Project\Weekend AI Projects\09-public-ship-log</code>

Telegram

<code>/log what you did 2h</code>	Log today's work (hours optional: 2h, 1.5 hours, 45m)
<code>/streak</code> · <code>/today</code>	Numbers · today's entries
<code>/undo</code> · <code>/draft</code>	Remove last /log · write today's post now
9 PM · 10:30 PM IST	Reminder (if nothing logged) · draft with Approve / Skip / Rewrite

Terminal

Tests	<code>C:\Python314\python -m unittest discover -s tests -t .</code>
Streak numbers	<code>C:\Python314\python -m shiplog status</code>
Pull commits now	<code>C:\Python314\python -m shiplog sync</code>
Rebuild this manual	<code>node docs\manual\build-docs.mjs</code> , then <code>docs\manual\print-pdfs.ps1</code>

Publish a change

```
cd "D:\Swapnil\Personel Project\Weekend AI Projects\09-public-ship-log"; git pull --rebase; git add -A; git commit -m "describe the change"; git push
```

Remember

Cost is ₹0. Everything you log is public. Nothing is posted until you press Approve and `BLUESKY_DRY_RUN` is 0. Run the bot in one place only.